

Felix Leitner

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Motivated and curious computer science student eager to pursue a career in software engineering, possessing a broad skill set that includes symbolic AI, languages, and algorithms. Currently a senior on track to receive a Master's degree in Spring 2026.

WORK EXPERIENCE

MathWorks, Cambridge, UK — *Development Intern*

JUNE 2025 - AUGUST 2025

- Developed a long-term testing tool for parallel computing clusters, targeting non-functional issues like memory drift and increasing execution time
- Continued work on adapting HPC schedulers to a containerised workflow and expanded documentation surrounding this area

Savant Systems, Hyannis, MA — *Development Intern*

JUNE 2024 - AUGUST 2024

- Adapted an existing open-source API integration testing tool to company requirements for use in quality assurance, saving 4 weeks of development time over re-implementation
- Redesigned and implemented the main page of a diagnostics tool used by installers and internal developers, improving user experience for viewing product information

TrackX Technology, Hillsborough, NC — *Development Intern*

JULY 2023 - AUGUST 2023

- Improved an internal .NET tool for managing remote product deployments, which included cryptographic key exchange and data encoding
- Created an expandable Python tool for exploring and visualizing SQLite datasets

EDUCATION

University of St Andrews, Scotland — Computer Science MSci

SEPTEMBER 2022 - APRIL 2026

- Started with second-year-level courses (direct entry)
- Projected for a First Class Honours degree
- Recognised on the Dean's List for academic achievement in 2023-2025

Classical High School, Providence, RI (USA)

SEPTEMBER 2019 - JUNE 2022

- 99.33 GPA - Ranked 6 / 260

OPEN SOURCE

[Conjure Oxide: Solver-Agnostic Constraint Programming Tool](#)

- Part of the core initial team building a tool for solving expressive constraint problems
- Led initial contributions to the core tree-rewriting engine, now being continued in current students' final year dissertations
- Adding an alternative SMT solver backend as part of my MSci dissertation

[Uniplate: Library for Generic Tree Traversal](#)

- Integrated the library with core language constructs
- Added several utility classes useful for downstream projects like Conjure Oxide

[Planning: A Generic Goal-Oriented Action Planning Library](#)

- Improved interface over existing libraries, supporting arbitrary state and cost heuristics

SKILLS

Native speaker in spoken and written German (Goethe Institut C1 certified)

Technical communication – experienced in making technical concepts approachable

Committed team player – skilled in solving complex issues alongside colleagues

Commitment to quality – focuses on creating reliable and user-friendly solutions

Expertise in front-end design/development as well as back-end architectural design

- Git, VS Code, Unix, Web, LaTeX

PROGRAMMING LANGUAGES

Python	8 years
HTML / CSS	6 years
Javascript	5 years
Rust	3 years
Java	3 years
C	2 years
Haskell	2 years
MATLAB	2 years
C#	1 year

COURSEWORK

- Compiler from a custom functional language to Java — Haskell, Java

- Implementing and comparing search algorithms for a flight route planner — Java

- Software engineering team project — Flask, React, TypeScript

- Implementing and evaluating the Paxos distributed consensus protocol — Java

- Reliable UDP file sharing application with loss recovery — Java

- Several projects using test-driven development

- Interpretive calculator language with file execution - Haskell

- Pipeline implementation using a custom thread-safe queue — C, Valgrind

- Gomoku AI using Minimax with aggressive move selection heuristic — Haskell, Gloss

LEADERSHIP

President (2024-26), Treasurer (2023-24), and Social Media Officer (2022-23) of the University of St Andrews Folk & Traditional Music Society

Taught for 3 years as a sailing instructor at West Dennis Yacht Club, Cape Cod

Mentored and assessed high school students in debate skills as a varsity debater